

□ Üretilen Peptidler - Optimizasyon Yöntemlerine Göre Özet

Oluşturulma Tarihi: Ocak 2026

Toplam Plastik Tipi: 6 (PET, PP, PE, PS, PVC, Nylon)

Toplam Peptid Sayısı: 180 (her plastik için 3 yöntem × 10 peptid)

□ Genel Performans Özeti

Plastik	SA En İyi	ILS En İyi	MOCO-CEM En İyi	Test R ²
PET	-63.14	-66.11	-65.35	0.9751
PP	-53.48	-55.44	-55.53	0.9593
PE	-59.59	-60.77	-59.62	0.9543
PS	-44.82	-44.90	-45.34	0.9583
PVC	-69.17	-68.86	-69.17	0.9489
Nylon	-77.71	-78.79	-78.79	0.9689

□ 1. Simulated Annealing (SA) Peptidleri

1.1 PET - SA Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	HHRWWMSQFWFH	-63.14	YQNRVSGAHGFW	-10.30	+52.84
2	HSRGGHGRVVDW	-62.51	YMYVARRMGQEK	-14.14	+48.37
3	FVWRWHLIMWVH	-59.57	YIDDVNTSMYIM	-10.41	+49.16
4	YMWWWWWQLRRRM	-59.46	AFQMFNTATRIE	-12.04	+47.42
5	WWFHRWNMFWKW	-59.37	QIIFGWQNFSEI	-10.75	+48.62
6	EYRAGGGRTTHKL	-57.81	VHQKWLGGNGNW	-11.22	+46.59
7	WKSVMRWQWLWI	-57.49	QAGGSRWDKAFM	-15.49	+42.00
8	FYWLRLMWWMR	-56.22	LYVGMFVWHQNE	-16.67	+39.55
9	TTWRMMIKMHSH	-53.34	KFYMDADQTKDI	-8.80	+44.54
10	RYEMMMRTWRYA	-51.04	TQVNRAGHDGDH	-8.89	+42.14

1.2 PP - SA Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WWQRHMFVWFHW	-53.48	YQNRVSGAHGFW	-10.55	+42.93
2	WWFRKMFVWFRW	-53.38	QAGGSRWDKAFM	-17.08	+36.29
3	WWMRVFMFFWRW	-51.97	LYVGMFVWHQNE	-16.75	+35.22
4	FWWHRIWTKRAW	-50.74	QIIFGWQNFSEI	-14.22	+36.52
5	WFQRWFIMFWRW	-49.59	TQVNRAGHDGDH	-4.28	+45.31
6	WWQWTHFYRYLW	-48.98	KFYMDADQTKDI	-6.20	+42.78
7	WRWWWRMHFRFW	-48.59	VHQKWLGGNGNW	-9.99	+38.60
8	WFWWFIAEVRIF	-46.38	YIDDVNTSMYIM	-9.88	+36.50
9	KYWHVRRWWRIY	-46.25	YMYVARRMGQEK	-9.45	+36.80

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
10	KGGMMWIKRQRM	-39.73	AFQMFNTATRIE	-6.67	+33.06

1.3 PE - SA Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	RWMWGFKWHMRH	-59.59	QAGGSRWKAFM	-25.25	+34.34
2	RWMWGFKWHMRH	-59.59	QIFGWQNFSEI	-17.99	+41.60
3	RWMWGFKWHMRH	-59.59	KFYMDADQTKDI	-9.33	+50.26
4	HWMWAMKWHMRH	-59.53	YIDDVNTSMYIM	-10.52	+49.01
5	RWHFRWMWHMRH	-58.17	YMYVARRMGQEK	-14.92	+43.25
6	VWRRLWRWRHRH	-58.14	TQVNRAGHDGDH	-7.81	+50.33
7	HFMLFIFWNMRW	-56.07	LYVGMFWVHQNE	-16.04	+40.03
8	WAWHWSRWVEFK	-53.20	VHQKWLGGNGNW	-15.48	+37.72
9	WMWIERRWHMLH	-51.46	AFQMFNTATRIE	-9.38	+42.08
10	RRAFHWWEHQWR	-49.78	YQNRVSGAHGFW	-15.75	+34.03

1.4 PS - SA Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WGMRWVFIHQM	-44.82	YIDDVNTSMYIM	-7.38	+37.44
2	AHRLWWQWHERM	-44.53	QIFGWQNFSEI	-8.89	+35.64
3	WWIRRGIWQGMR	-43.90	LYVGMFWVHQNE	-12.01	+31.89
4	RKRMWWQWHHRM	-43.76	TQVNRAGHDGDH	-9.43	+34.32
5	RLFQWWQWHFRM	-43.70	KFYMDADQTKDI	-5.94	+37.76
6	WWWWWQLRFWFR	-42.52	YMYVARRMGQEK	-8.27	+34.25
7	WMRWWQLRMWFR	-42.18	VHQKWLGGNGNW	-5.34	+36.84
8	WFRWWIDQRWFR	-40.99	QAGGSRWKAFM	-9.00	+31.99
9	WMFRHMLWHVMR	-38.57	YQNRVSGAHGFW	-6.51	+32.05
10	WKWRFWIHRLW	-36.59	AFQMFNTATRIE	-10.49	+26.11

1.5 PVC - SA Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	HWNWIQNFMFIR	-69.17	VHQKWLGGNGNW	-25.59	+43.58
2	WWHWRNNFQFYR	-68.59	KFYMDADQTKDI	-25.23	+43.36
3	HWVWQQNFMRIR	-68.42	AFQMFNTATRIE	-25.43	+42.99
4	WWRWQNQMFFIR	-66.92	YIDDVNTSMYIM	-26.17	+40.76
5	WWWWSNEERQR	-65.21	QAGGSRWKAFM	-19.25	+45.96
6	RFWWQTRVHRIW	-63.12	YQNRVSGAHGFW	-23.11	+40.01
7	RMISFRKL RVV	-60.52	QIFGWQNFSEI	-22.51	+38.01
8	GMLFYRQLRKWN	-59.64	LYVGMFWVHQNE	-27.57	+32.07
9	VMMIWRILRQH	-59.29	YMYVARRMGQEK	-27.63	+31.66
10	WWQWRREWWML	-58.72	TQVNRAGHDGDH	-22.46	+36.26

1.6 Nylon - SA Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WWRMRDWWLRWW	-77.71	YIDDVNTSMYIM	-27.16	+50.55
2	WWRMRDWWLRWW	-77.71	YMYVARRMGQEK	-32.24	+45.47
3	MWWWMSWWMRWW	-77.52	QAGGSRWDKAFM	-21.28	+56.24
4	FWWMRDWRLRWW	-76.81	TQVNRAGHDGDH	-23.17	+53.64
5	WWYMRQWRRWW	-76.68	QIIFGWQNFSEI	-23.44	+53.24
6	MWEMRQWFLRQW	-76.12	VHQKWLGGNGNW	-23.84	+52.27
7	KFYFRYWRRWR	-72.31	LYVGMFWVHQNE	-29.23	+43.08
8	WRRKWRYWEEWR	-69.93	YQNRVSGAHGFW	-28.17	+41.77
9	WRMHWRMWNEW	-69.42	AFQMFNTATRIE	-28.54	+40.89
10	WRHYWMYMVEWR	-67.67	KFYMDADQTKDI	-29.28	+38.40

□ 2. Iterated Local Search (ILS) Peptidleri

2.1 PET - ILS Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WEWWFGFHHRLR	-66.11	VHQKWLGGNGNW	-11.22	+54.89
2	WEWWFVFHHRLM	-65.58	KFYMDADQTKDI	-8.80	+56.79
3	WEWWFVFHHRLM	-65.58	LYVGMFWVHQNE	-16.67	+48.91
4	WEWWAHFHHRLR	-64.18	YMYVARRMGQEK	-14.14	+50.04
5	YWIFQHWRWMAR	-63.28	QAGGSRWDKAFM	-15.49	+47.79
6	YWKLLHWRWMAR	-63.05	TQVNRAGHDGDH	-8.89	+54.16
7	HGRGGHGRFGLW	-62.33	YIDDVNTSMYIM	-10.41	+51.92
8	MWFHMWTFQFRW	-61.70	AFQMFNTATRIE	-12.04	+49.66
9	WWQRSYHHWWWG	-60.20	YQNRVSGAHGFW	-10.30	+49.91
10	RWQRYHNNWWWE	-59.99	QIIFGWQNFSEI	-10.75	+49.24

2.2 PP - ILS Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WWQRHYFFWRMW	-55.44	TQVNRAGHDGDH	-4.28	+51.16
2	WWQRHYFFWRAW	-55.40	YMYVARRMGQEK	-9.45	+45.94
3	WWQRHKFFWRMW	-54.71	LYVGMFWVHQNE	-16.75	+37.96
4	WWQRHKFFFWMW	-54.44	YIDDVNTSMYIM	-9.88	+44.56
5	WWQRHKFFFWMW	-54.44	YQNRVSGAHGFW	-10.55	+43.89
6	WWWRRREYFWMW	-54.36	QIIFGWQNFSEI	-14.22	+40.14
7	WWQRQFFWRMW	-54.33	QAGGSRWDKAFM	-17.08	+37.24
8	WWERREFFWRMW	-54.25	VHQKWLGGNGNW	-9.99	+44.27
9	WWFRKMFQWFHW	-53.52	KFYMDADQTKDI	-6.20	+47.33
10	WWHHKMFQWFHW	-52.80	AFQMFNTATRIE	-6.67	+46.13

2.3 PE - ILS Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	AHYHFLWHQMRW	-60.77	AFQMFNTATRIE	-9.38	+51.38
2	KWMWKMKWWMRH	-59.62	VHQKWLGGNGNW	-15.48	+44.15
3	RWMWAMKWHLRH	-59.56	YMYVARRMGQEK	-14.92	+44.65
4	HWMWNMKWQMRH	-58.73	YIDDVNTSMYIM	-10.52	+48.21
5	KWMWWMKWEMRL	-58.72	TQVNRAHGDGDH	-7.81	+50.90
6	WWRRFWRWHHFFH	-58.51	QAGGSRWDKAFM	-25.25	+33.26
7	WWRRFWRWHHFFH	-58.51	QIIFGWQNFSEI	-17.99	+40.51
8	SWMWHFKWWMRH	-58.29	LYVGMFWVHQNE	-16.04	+42.25
9	WWRRFWRWLHFFH	-58.23	KFYMDADQTKDI	-9.33	+48.90
10	FWMRGWRWHHRH	-57.56	YQNRVSGAHGFW	-15.75	+41.81

2.4 PS - ILS Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WWHIRNIWQEMR	-44.90	TQVNRAHGDGDH	-9.43	+35.46
2	WWWIRVIWQHMR	-44.90	YIDDVNTSMYIM	-7.38	+37.52
3	WHWQRGIWQGMR	-44.73	LYVGMFWVHQNE	-12.01	+32.72
4	WWWIRNIWQQMR	-44.65	AFQMFNTATRIE	-10.49	+34.17
5	WWWQRVIWQHMR	-44.64	YMYVARRMGQEK	-8.27	+36.37
6	WWRWRIIWQHMR	-44.40	QIIFGWQNFSEI	-8.89	+35.51
7	WHWYRFIWQQMR	-44.28	VHQKWLGGNGNW	-5.34	+38.94
8	WGMRWHTVLHQM	-44.04	KFYMDADQTKDI	-5.94	+38.10
9	WWWQWQLRFWFR	-42.72	YQNRVSGAHGFW	-6.51	+36.21
10	WWWQWIHRFWFR	-42.56	QAGGSRWDKAFM	-9.00	+33.55

2.5 PVC - ILS Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WWIWIQNFMIR	-68.86	TQVNRAHGDGDH	-22.46	+46.40
2	WWNWIQNFMRIR	-68.67	YQNRVSGAHGFW	-23.11	+45.56
3	HWNWLQNFMIRIR	-68.32	LYVGMFWVHQNE	-27.57	+40.75
4	WWDWRHNFFIR	-67.86	QIIFGWQNFSEI	-22.51	+45.35
5	HWVWQNNFMIRIR	-67.09	QAGGSRWDKAFM	-19.25	+47.84
6	RWYQNVWRMRIF	-66.33	YMYVARRMGQEK	-27.63	+38.70
7	RWYNLQRWMRIW	-65.85	KFYMDADQTKDI	-25.23	+40.61
8	RHMYRWILRLWQ	-63.65	AFQMFNTATRIE	-25.43	+38.22
9	RHMIRYQLRLWV	-62.70	YIDDVNTSMYIM	-26.17	+36.54
10	RHMYQRHLRLWS	-62.70	VHQKWLGGNGNW	-25.59	+37.11

2.6 Nylon - ILS Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	YWRMMDWWLRWW	-78.79	TQVNRAHGDGDH	-23.17	+55.62
2	YWRMMDWWLRWW	-78.79	KFYMDADQTKDI	-29.28	+49.51
3	YWRMMDWWLRWW	-78.79	YQNRVSGAHGFW	-28.17	+50.62
4	YWRMMDWWLRWW	-78.79	QIIFGWQNFSEI	-23.44	+55.35

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
5	YWRMMDWWLRWW	-78.79	YMYVARRMGQEK	-32.24	+46.55
6	YWRMMNWWLRWW	-78.28	VHQQWLGGNGNW	-23.84	+54.44
7	MWRMMSWWLRWW	-78.17	LYVGMFWVHQNE	-29.23	+48.94
8	MWRMMSWWLRWW	-78.17	QAGGSRWDKAFM	-21.28	+56.89
9	YWRMRDWWLRWW	-77.61	AFQMFNTATRIE	-28.54	+49.07
10	RWWLMAWWRRWW	-76.87	YIDDVNTSMYIM	-27.16	+49.72

□ 3. MOCO-CEM Peptidleri

3.1 PET - MOCO-CEM Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	YGRGGWGRQEWW	-65.35	AFQMFNTATRIE	-12.04	+53.31
2	YGRGGWGRQEWW	-65.35	YIDDVNTSMYIM	-10.41	+54.94
3	YGRGGWGRQEWW	-65.35	TQVNRAHDGDH	-8.89	+56.45
4	YGRGGWGRQEWW	-65.35	QIIFGWQNFSEI	-10.75	+54.60
5	YGRGGWGRQEWW	-65.35	YQNRVSGAHGFW	-10.30	+55.05
6	YGRGGWGRQEWW	-65.35	YMYVARRMGQEK	-14.14	+51.21
7	WWRRMWYWRWWR	-60.57	KFYMDADQTKDI	-8.80	+51.78
8	WWQRSYHHWWWG	-60.20	QAGGSRWDKAFM	-15.49	+44.71
9	WWQRSYHHWWWG	-60.20	VHQQWLGGNGNW	-11.22	+48.98
10	WWQRSYHHWWWG	-60.20	LYVGMFWVHQNE	-16.67	+43.53

3.2 PP - MOCO-CEM Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WWQRRAFFWRMW	-55.53	QAGGSRWDKAFM	-17.08	+38.44
2	WWQRRAFFWRMW	-55.53	AFQMFNTATRIE	-6.67	+48.85
3	WWQRRAFFWRMW	-55.53	YIDDVNTSMYIM	-9.88	+45.65
4	WWQRRAFFWRMW	-55.53	TQVNRAHDGDH	-4.28	+51.25
5	WWQRRAFFWRMW	-55.53	QIIFGWQNFSEI	-14.22	+41.30
6	WWQRRAFFWRMW	-55.53	YQNRVSGAHGFW	-10.55	+44.97
7	WWQRRAFFWRMW	-55.53	VHQQWLGGNGNW	-9.99	+45.54
8	WWQRRAFFWRMW	-55.53	KFYMDADQTKDI	-6.20	+49.33
9	WWQRRAFFWRMW	-55.53	YMYVARRMGQEK	-9.45	+46.07
10	WWQRRAFFWRMW	-55.53	LYVGMFWVHQNE	-16.75	+38.77

3.3 PE - MOCO-CEM Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	KWMWKMKWHMRH	-59.62	QAGGSRWDKAFM	-25.25	+34.37
2	KWMWKMKWHMRH	-59.62	AFQMFNTATRIE	-9.38	+50.24
3	KWMWKMKWHMRH	-59.62	YIDDVNTSMYIM	-10.52	+49.10

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
4	KWMWKMKWHRH	-59.62	TQVNRAGHDGDH	-7.81	+51.81
5	KWMWKMKWHRH	-59.62	QIIFGWQNFSEI	-17.99	+41.63
6	KWMWKMKWHRH	-59.62	YQNRVSGAHGFW	-15.75	+43.87
7	KWMWKMKWHRH	-59.62	VHQQWLGGNGNW	-15.48	+44.15
8	KWMWKMKWHRH	-59.62	YMYVARRMGQEK	-14.92	+44.71
9	KWMWKMKWHRH	-59.62	LYVGMFWVHQNE	-16.04	+43.58
10	HWMWAMKWAMRH	-59.10	KFYMDADQTKDI	-9.33	+49.77

3.4 PS - MOCO-CEM Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	WHWQREIWQSMR	-45.34	QAGGSRWDKAFM	-9.00	+36.34
2	WHWQREIWQSMR	-45.34	AFQMFNTATRIE	-10.49	+34.86
3	WHWQREIWQSMR	-45.34	YIDDVNTSMYIM	-7.38	+37.97
4	WHWQREIWQSMR	-45.34	YQNRVSGAHGFW	-6.51	+38.83
5	WHWQREIWQSMR	-45.34	LYVGMFWVHQNE	-12.01	+33.34
6	WHWQREIWQSMR	-45.34	YMYVARRMGQEK	-8.27	+37.07
7	WHWIRGIWQGMR	-45.27	TQVNRAGHDGDH	-9.43	+35.84
8	WHWIRGIWQGMR	-45.27	QIIFGWQNFSEI	-8.89	+36.38
9	WHWIRGIWQGMR	-45.27	VHQQWLGGNGNW	-5.34	+39.94
10	WWWQWQLRFWFR	-42.72	KFYMDADQTKDI	-5.94	+36.78

3.5 PVC - MOCO-CEM Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	HWNWIQNFMFIR	-69.17	QAGGSRWDKAFM	-19.25	+49.92
2	HWNWIQNFMFIR	-69.17	AFQMFNTATRIE	-25.43	+43.74
3	HWNWIQNFMFIR	-69.17	YIDDVNTSMYIM	-26.17	+43.00
4	HWNWIQNFMFIR	-69.17	TQVNRAGHDGDH	-22.46	+46.71
5	HWNWIQNFMFIR	-69.17	QIIFGWQNFSEI	-22.51	+46.66
6	HWNWIQNFMFIR	-69.17	YQNRVSGAHGFW	-23.11	+46.06
7	HWNWIQNFMFIR	-69.17	VHQQWLGGNGNW	-25.59	+43.58
8	HWNWIQNFMFIR	-69.17	KFYMDADQTKDI	-25.23	+43.93
9	HWNWIQNFMFIR	-69.17	YMYVARRMGQEK	-27.63	+41.54
10	HWNWIQNFMFIR	-69.17	LYVGMFWVHQNE	-27.57	+41.60

3.6 Nylon - MOCO-CEM Peptidleri

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
1	YWRMMDWWLRWW	-78.79	QAGGSRWDKAFM	-21.28	+57.51
2	YWRMMDWWLRWW	-78.79	AFQMFNTATRIE	-28.54	+50.25
3	YWRMMDWWLRWW	-78.79	YIDDVNTSMYIM	-27.16	+51.63
4	YWRMMDWWLRWW	-78.79	TQVNRAGHDGDH	-23.17	+55.62
5	YWRMMDWWLRWW	-78.79	QIIFGWQNFSEI	-23.44	+55.35
6	YWRMMDWWLRWW	-78.79	YQNRVSGAHGFW	-28.17	+50.62
7	YWRMMDWWLRWW	-78.79	VHQQWLGGNGNW	-23.84	+54.95

#	Üretilen Peptid	Skor	Başlangıç Peptid	Baş. Skor	İyileştirme
8	YWRMMDWWLRWW	-78.79	KFYMDADQTKDI	-29.28	+49.51
9	YWRMMDWWLRWW	-78.79	YMYVARRMGQEK	-32.24	+46.55
10	YWRMMDWWLRWW	-78.79	LYVGMFWVHQNE	-29.23	+49.56

□ 4. Yöntem Karşılaştırması

4.1 En İyi Benzersiz Peptidler (Tüm Yöntemler)

Plastik	En İyi Peptid	Skor	Yöntem
Nylon	YWRMMDWWLRWW	-78.79	ILS / MOCO-CEM □
PVC	HWNWIQNFMFIR	-69.17	SA / MOCO-CEM
PET	WEWWFGFHHRLR	-66.11	ILS
PE	AHYHFLWHQMRW	-60.77	ILS
PP	WWQRRAFFWRMW	-55.53	MOCO-CEM
PS	WHWQREIWQSMR	-45.34	MOCO-CEM

4.2 Yöntem Bazında Ortalama Performans

Yöntem	Ort. En İyi Skor	En İyi Plastik	En Kötü Plastik
SA	-60.67	Nylon (-77.71)	PS (-44.82)
ILS	-62.31	Nylon (-78.79)	PS (-44.90)
MOCO-CEM	-62.30	Nylon (-78.79)	PS (-45.34)

4.3 Çeşitlilik Analizi

Yöntem	Benzersiz Peptid Oranı	Açıklama
SA	~95%	Yüksek çeşitlilik
ILS	~85%	Orta çeşitlilik
MOCO-CEM	~40%	Düşük çeşitlilik (convergence)

□ 5. Amino Asit Motif Analizi

5.1 En Sık Görülen Başlangıç Motifleri

Motif	Frekans	Plastikler
WW	45%	Tüm plastikler
HW	20%	PVC, PE
YW	15%	Nylon
KW	10%	PE

5.2 En Sık Görülen Bitiş Motifleri

Motif	Frekans	Plastikler
RW	35%	Nylon, PET
RH	25%	PE
IR	20%	PVC
MR	15%	PS

Not: Bu rapor peptide_generation_comparison.py scriptinin çıktılarından otomatik olarak derlenmiştir.